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Swift or Exact? Boosting Efficient Microarchitecture DSE via Multi-fidelity Partial Order Prediction

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***Abstract*—A significant challenge in microarchitecture design space exploration (DSE) lies in the time-intensive synthesis and simulation process, making rapid design exploration infeasible. While the simulation tools offer reports on performance, power, and area (PPA) in the different stages, the PPA reports at early stages may fail to reflect the true relative qualities for various designs, *i.e.*, with low fidelities. To address these limitations, we propose a novel multi-fidelity optimization algorithm tailored for multi-stage optimization problems. The proposed method employs a non-linear Gaussian process model to effectively fuse data from different stages with different fidelities, minimizing the need for expensive high-fidelity data while maximizing accuracy. Furthermore, a logical regression function and a multi-objective partial order relation are introduced to evaluate the reliability of low-fidelity data, mitigating their potential inaccuracies. Experiments demonstrate that our proposed multi-fidelity optimization algorithm can approximate the Pareto front of the direct design space in a shorter time with better performance.**

I. Introduction

The rapid advancement of artificial intelligence algorithms places increasingly stringent demands on hardware design [1]. Developing a high-quality microarchitecture typically requires specialized expertise and extensive engineering efforts, often resulting in prolonged development cycles. Recently, academia and industry have proposed and adopted systematic, efficient, and universal development processes to mitigate design overhead and expedite design cycles [2]–[4]. For instance, platforms like Chipyard [5] and Gem5 [6] use configurable modules and design parameters to transform intricate microarchitecture design optimization challenges into essential design space exploration problems.

However, the intricate nature of the very large-scale integration (VLSI) flow continues to hinder efficient microarchitecture design. The process encompasses architecture design, register-transfer level (RTL) design, netlist design stages, *etc.*, resulting in a time-consuming workflow. While early-stage performance metrics enable cross-layer optimization [7], [8], the low accuracy of these reports poses a significant risk of missing optimal solutions, as they do not correspond to the final, more accurate results.

Instead of focusing on accurately predicting the values of PPA [9], [10], our primary concern is whether the predictions at different stages can reliably reflect the true relative performance ranking among designs, which we define as fidelity. In the microarchitecture development flow, data from different stages with varying fidelity levels are available. Therefore, we

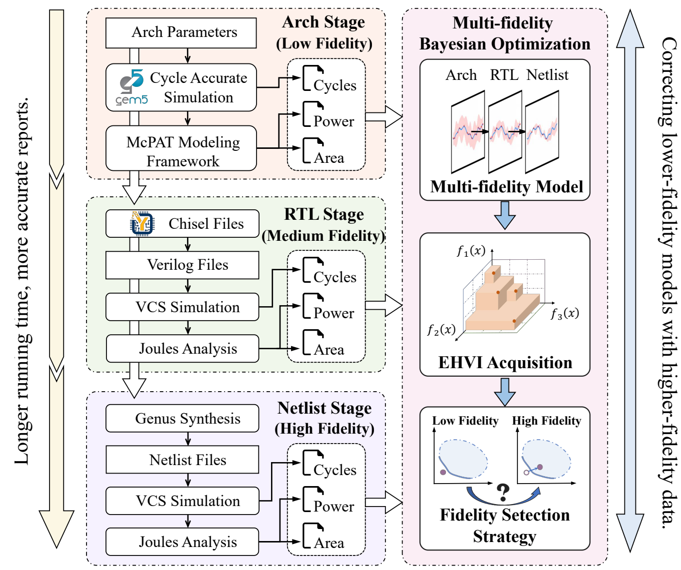

Fig. 1 Flow of agile microarchitecture development with multi-fidelity analytical reporting. Higher-fidelity data are used to correct lower-fidelity models.

introduce multi-fidelity optimization into microarchitecture DSE to further enhance the efficiency of the exploration process, as shown in Fig. 1. The core challenges of multi-fidelity optimization lie in effectively utilizing multi-fidelity data and determining the appropriate fidelity level for the next evaluation.

Early methods often assumed that the relationships between different fidelity levels were independent or linear [11], [12]. However, in microarchitecture design, reports across different fidelity levels often exhibit non-linear relationships. To address this challenge, Zhang *et al.* [13] use Gaussian models to capture non-linear mappings between low-fidelity and high-fidelity data and optimize on the fused model. Sun *et al.* [14] extend this to multi-objective problems with a correlated multi-objective non-linear optimization algorithm for exploring Pareto solutions using multi-fidelity data. Daulton *et al.* [15] incorporate fidelity level as a new parameter feature, using a knowledge gradient-based general one-step lookahead function for simultaneous optimization. Irshad *et al.* [16] enhance this with a multi-objective policy that includes trust gain per evaluation cost, enabling efficient multi-objective, multi-fidelity optimization.

Unfortunately, these methods fail to leverage the interpretability of data from different fidelities. If low-fidelity data accurately ranks candidate solutions, additional high-fidelity evaluations become unnecessary. High-fidelity evaluations should only be conducted when low-fidelity data cannot reliably rank exploration points. Additionally, defining an effective partial order for different points is essential, especially when ranking complex multi-objective problems across fidelity levels.

In this work, to efficiently utilize multi-fidelity data, we propose an efficient multi-fidelity Bayesian optimization framework based on a non-linear multi-fidelity model and partial order prediction, as shown in the Fig. 1. Our contributions are summarized as follows:

- We utilize non-linear Gaussian modeling to integrate data of varying fidelities and employ the expected hypervolume improvement (EHVI) acquisition function to balance exploitation and exploration.
- We employ logistic regression to predict the potential inversion of PPA values between target points and current Pareto points across different fidelity levels.
- A decision-making framework has been proposed to evaluate the partial order relationship in multi-objective problems, facilitating the efficient discovery of new Pareto points.
- We conduct comprehensive experiments on parameterized RISC-V CPUs, benchmarked on a widely recognized RISC-V benchmark suite. Results demonstrate that our method increases hypervolume by 12.9%, enhances Pareto front quality by 57.7%, and achieves a 48.0% faster convergence speed compared to the previous method.

II. Preliminaries

A. RISC-V CPU

The RISC-V architecture, a reduced instruction set computer architecture, has gained widespread adoption in recent years due to its efficient hardware design and implementation, coupled with its remarkable power efficiency [17]. Frameworks such as Chipyard [5] further accelerate the development of RISC-V CPUs by providing an open-source platform for integrating diverse components. Chipyard integrates various RISC-V CPUs, including the high-performance Berkeley Out-of-Order Machine (BOOM) [18] and the streamlined, efficient Rocket [19] in-order CPU.

B. Multi-objective Optimization

Bayesian optimization [20] is a powerful global optimization method for black-box functions with high evaluation costs, such as microarchitecture design. It approximates the objective function with a surrogate model, often a Gaussian process, and uses an acquisition function to balance exploration and exploitation.

Optimizing microarchitecture designs often requires balancing multiple, conflicting objectives. Multi-objective Bayesian optimization (MOBO) extends standard Bayesian optimization, allowing efficient exploration of multi-objective design space. Let functions $\{f_i(\boldsymbol{x})\}_{i=1}^{m}$, *i.e.*, f_1 = power, f_2 = area, f_3 = cycles, denote the m-dimension metrics to be minimized and $\mathbb{X}$ denotes the parameter space. A parameter vector $\boldsymbol{x}^* \in \mathbb{X}$ is said to dominate $\boldsymbol{x}' \in \mathbb{X}$, denoted as $\boldsymbol{x}^* \succeq \boldsymbol{x}'$, if the following two conditions are met in this minimization problem:

1) $f_i(\boldsymbol{x}') \geq f_i(\boldsymbol{x}^*), \forall i \in \{1, ..., m\}$,
2) There exists at least one j such that $f_j(\boldsymbol{x}') > f_j(\boldsymbol{x}^*)$.

The set of parameter vectors that are not dominated by other vectors is called the Pareto-optimal set, denoted as $\mathbb{X}_{Pareto}$, which is the target of MOBO:

$$\mathbb{X}_{Pareto} = \{\boldsymbol{x}^* \in \mathbb{X} | \nexists \boldsymbol{x}' \in \mathbb{X}, \boldsymbol{x}' \succeq \boldsymbol{x}^*\}. \tag{1}$$

For the problem with multiple optimization objectives, Bayesian optimization constructs surrogate models for these unknown black-box objectives. A multi-objective acquisition function is then constructed based on these surrogate models to evaluate the quality of a parameter configuration $\boldsymbol{x}$ for identifying the Pareto set, eliminating the need for extensive and time-consuming EDA flow evaluations.

C. Multi-fidelity Optimization

Fidelity refers to the degree to which a model accurately represents the state of a real-world project or application. It essentially quantifies the realism of the model. Low-fidelity models are characterized by lower precision, while high-fidelity models exhibit higher precision.

For a multi-fidelity problem, each fidelity i has a regression model $f_i(\boldsymbol{x})$. The multi-fidelity model can be defined as:

$$f_{i+1}(\boldsymbol{x}) = z(f_i(\boldsymbol{x}), \boldsymbol{x}) \tag{2}$$

where $z(\cdot)$ is a machine learning model or function, f_i is the model of the lower fidelity and f_{i+1} is the higher one. Higher fidelity levels offer greater accuracy but come at the cost of significantly longer running times compared to lower fidelity levels. If low-fidelity results are satisfactory, we can skip higher-fidelity tests, to save time. Therefore, we need to measure the quality of reports at each fidelity level to determine if subsequent design stages are necessary. In the rest of this paper, the three fidelities in microarchitecture design shown in Fig. 1 are shorted as $arch$, rtl, and $netlist$.

D. Problem Formulation

Definition 1 (Hypervolume)**.** In multi-objective problems, solutions are defined in terms of Pareto fronts. The quality of such fronts can be measured by hypervolume [21]:

$$\mathrm{HV}\left(f^{\mathrm{ref}}, \mathbb{X}\right) = \Lambda\left(\bigcup_{\mathbb{X}_n \in \mathbb{X}} \left[f_1(\mathbb{X}_n), f_1^{\mathrm{ref}}\right] \times \cdots \times \left[f_m(\mathbb{X}_n), f_m^{\mathrm{ref}}\right]\right), \tag{3}$$

where f^{ref} denotes a chosen reference point, typically set to represent a very poor performance value. f_m means the m-th metric, *i.e.*, cycles, power, and area. Λ refers to the Lebesgue measure.

Definition 2 (ADRS)**.** Another metric to measure the quality of Pareto fronts is the average distance to reference set (ADRS) [22]:

$$\mathrm{ADRS}(\Gamma, \Omega) = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \min_{\omega \in \Omega} D(\gamma, \omega), \tag{4}$$

where D is the Euclidean distance function. Γ is the real Pareto-optimal set and Ω is the learned Pareto-optimal set. ADRS is often exploited to measure how close a learned Pareto-optimal set is to the real Pareto-optimal set of the design space.

Problem 1 (Multi-fidelity microarchitecture DSE)**.** Given a search space $\mathbb{X}$, the metric space including three different fidelity levels: $\{\mathbb{Y}_{arch}, \mathbb{Y}_{rtl}, \mathbb{Y}_{netlist}\}$, we define the multi-fidelity microarchitecture DSE as finding the subset $\mathbb{X}^* \in \mathbb{X}$ with corresponding metrics $\mathbb{Y}^*_{netlist}$ forming the Pareto optimal set, by utilizing data from different fidelity levels.

III. Multi-fidelity Bayesian optimization

In this section, we present a multi-fidelity Bayesian optimization method designed to efficiently explore the microarchitecture design space by utilizing data from various fidelity levels with partial order prediction.

A. Overall Optimization Flow

Fig. 2 shows an overview of the proposed multi-fidelity Bayesian optimization. Firstly, we determine the design space to be explored based on the architecture constraints and engineers' expertise. Since the three fidelity stages are performed sequentially, the points run at a higher-fidelity stage are subsets of the points run at a lower-fidelity stage, *i.e*, $\mathrm{X}_{netlist} \subseteq \mathrm{X}_{rtl} \subseteq \mathrm{X}_{arch}$. These points are then fed into the VLSI flow to get corresponding fidelity reports. To fully leverage data from different fidelity stages, we construct and initialize a non-linear multi-fidelity Gaussian Process (GP) model. The expected hypervolume improvement acquisition function guides the selection of the next sampling point x^*, balancing the trade-off between exploration and exploitation. In each optimization time step, we employ a sequential fidelity strategy. After obtaining a fidelity report for x^* from the VLSI flow, the report is evaluated to determine if further evaluation at the next fidelity stage is necessary. The learned Pareto optimal set is acquired when a pre-determined time budget is met.

B. Non-linear Multi-fidelity Model

Generally, in multi-fidelity optimization, the relationships between multiple fidelities are assumed to be linear, *e.g.*, [12]. However, this assumption does not hold for the three stages of the VLSI flow, as they exhibit strong non-linear correlations. Therefore, we propose a non-linear multi-fidelity model [23] to further exploit the corresponding non-linear correlations between low-fidelity and high-fidelity data. The first level of the multi-fidelity model is based on the traditional GP model. Then the non-linear model can be formulated as Equation (5).

$$f_h(x) = z(f_l(x)) + \delta(x), \tag{5}$$

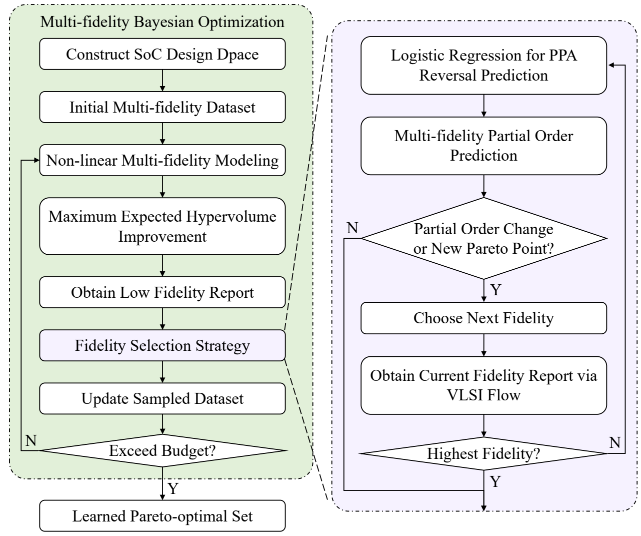

Fig. 2 Overview of the proposed multi-fidelity Bayesian optimization.

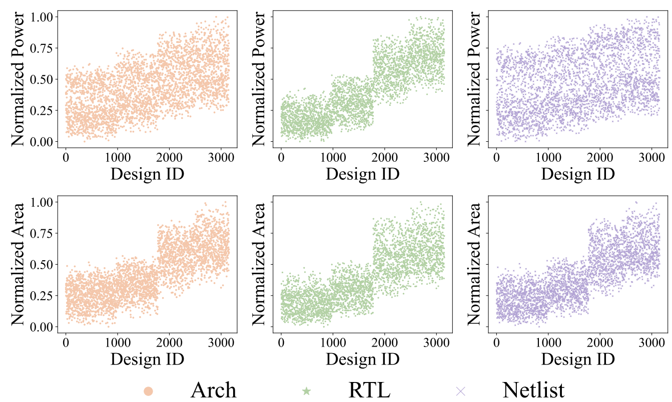

Fig. 3 Normalized power and area values of the BOOM design across three fidelity stages.

where $z(\cdot)$ represents the non-linear correlation between the low-fidelity and high-fidelity data, while $\delta(\cdot)$ is a Gaussian process model for the high-fidelity data, independent of the low-fidelity model. By concatenating the outputs of the low-fidelity model with the features of the architectural parameters as input features for the high-fidelity GP model, we can effectively fuse low-fidelity and high-fidelity information by constructing $GP(f_l(x), x)$.

The power and area values obtained at different fidelity stages are shown in Fig. 3 as examples to illustrate the complex non-linear relationships between the three fidelities. It can be seen that the reports from the three fidelities exhibit high divergence, making it difficult to regress their relationships using traditional linear models. Therefore, employing a non-linear model is a sensible choice for handling multi-fidelity exploration of microarchitecture design space.

C. Multi-fidelity Acquisition Function

After constructing a non-linear multi-fidelity model that integrates information from all fidelities, we introduce Expected

Hypervolume Improvement (EHVI) [24] as the acquisition function for efficient learning and measurement of the Pareto set in multi-objective optimization problems.

Assuming at the current optimization step $t+1$, we have a set of parameter configurations $\mathbb{X}_h = \{\boldsymbol{x}_h\}^t$ and their corresponding values $\mathbb{Y}_h = \{\boldsymbol{y}_h\}^t$ at the highest fidelity level. A virtual configuration point f^{ref} is a reference point, set to extremely large values for all design objectives in a multi-objective minimization problem. At each iteration, we aim to sample a parameter configuration that leads to the highest expected hypervolume improvement of the sampling configuration set. To achieve this, we will traverse the entire unsampled configuration space and estimate the expected hypervolume improvement for each configuration at the highest fidelity. The expected hypervolume improvement is defined as Equation (6).

$$\begin{aligned}&\text{EHVI}(\boldsymbol{x}_{t+1}|\mathbb{X},\mathbb{Y}) = \\ &\mathbb{E}_{p(\boldsymbol{y}(\boldsymbol{x}_{t+1})|\mathbb{X},\mathbb{Y})}\left[\text{HV}(f^{\text{ref}},\mathbb{Y}\cup\boldsymbol{y}(\boldsymbol{x}_{t+1}))-\text{HV}(f^{\text{ref}},\mathbb{Y})\right].\end{aligned} \tag{6}$$

By finding a parameter configuration that maximizes the EHVI, we can effectively balance exploration and exploitation in complex multi-objective optimization problems, facilitating the discovery of an improved Pareto set.

D. *Fidelity Selection Strategy*

In multi-fidelity optimization, in addition to employing an acquisition function to select the next sampling point, it is also crucial to determine the evaluation fidelity for the multi-fidelity Bayesian optimization process. Due to the significantly higher computational cost of high-fidelity simulations compared to low-fidelity ones, we should only sample high-fidelity data when necessary. The search process remains unaffected by varying fidelity data as long as all comparisons are accurately ranked. Given two parameter configurations with a significant difference in their low-fidelity metrics, the ranking based on high-fidelity data is unlikely to deviate from the ranking based on low-fidelity data. Therefore, we can skip the time-consuming high-fidelity simulation process. High-fidelity data is only necessary for more accurate assessments when low-fidelity data cannot confidently rank the target points. In multi-objective optimization, we aim to sample a new Pareto point in each optimization step, meaning the point that is not dominated by current Pareto points.

Therefore, we propose an adaptive fidelity selection strategy based on partial order prediction. This strategy utilizes a logistic regression function to predict the potential reversal of the PPA between the target point and the current Pareto set across different fidelity levels. In multi-objective optimization, we define the dominance of a newly sampled point by the current Pareto set as a partial order relationship. By combining the prediction results of the logistic regression function, we can predict the change in the partial order relationship between two points across low- and high-fidelity levels. This prediction informs the decision of whether or not to employ high-fidelity simulations.

Algorithm 1 Fidelity Selection Strategy

Input: The next sampling point x^*, current pareto metrics set $\mathbb{Y}_p$, Logistic regression function LR_i^m, where $i \in \{arch, rtl\}, m \in \{power, area, cycles\}$.

Output: The set of simulation results for x^* at different fidelity levels $\mathbb{Y}^*$.

1: Initialize $i \leftarrow arch$, $\boldsymbol{y}_i^* \leftarrow \text{VLSI}(x^*, i)$, and $\mathbb{Y}^* \leftarrow \boldsymbol{y}_i^*$;
2: **for all** $\boldsymbol{y}_p \in \mathbb{Y}_p$ **do**
3: Assess the partial ordering of the two points at the current fidelity:
$$PO_i \leftarrow (\boldsymbol{y}_p, \boldsymbol{y}_i^*);$$
4: Predict potential PPA reversals between current and highest fidelity using a logistic regression model:
$$REV_i = LR_i^m(\boldsymbol{y}_p, \boldsymbol{y}_i^*);$$
5: Predict the partial ordering of the two points at the highest fidelity:
$$PO_h \leftarrow REV_i;$$
6: **if** PO_h is not equal to PO_i **then**
7: $i = i+1$, $\boldsymbol{y}_i^* \leftarrow \text{VLSI}(x^*, i)$;
8: $\mathbb{Y}^* \leftarrow \mathbb{Y}^* \cup \boldsymbol{y}_i^*$, go back to line 2;
9: **end if**
10: **end for**
11: **if** $\boldsymbol{y}_i^*$ is a new Pareto metrics at fidelity i **then**
12: $i = i+1$, $\boldsymbol{y}_i^* \leftarrow \text{VLSI}(x^*, i)$;
13: $\mathbb{Y}^* \leftarrow \mathbb{Y}^* \cup \boldsymbol{y}_i^*$, go back to line 2;
14: **end if**
15: **return** $\mathbb{Y}^*$.

Algorithm 1 presents a detailed description of the fidelity selection strategy applied at each step of the optimization process. In line 1, we start by using the lowest fidelity level to obtain data for the newly sampled point within the VLSI process. Next, we sequentially compare the new point with all current Pareto points to evaluate the partial order at the current fidelity (line 3).

To infer the partial order relationship between the two points at high fidelity, we employ a logistic regression function to predict the potential reversal of their PPA values between low and high fidelity levels (line 4). After training on the previously sampled dataset, the relationship between the difference in the PPA values of the two points at $i-th$ fidelity $\Delta = |y_i^m - y_p^m|$ and the probability of a wrong ranking for the highest fidelity level can be described by a logistic regression model as Equation (7):

$$\text{LR}_i^m(y_i^m, y_p^m) = \frac{1}{1+e^{-(\beta_0+\beta_1\Delta)}}, \tag{7}$$

where the hyperparameters β_0 and β_1 are trained on the currently sampled high-fidelity data and its corresponding low-fidelity data. Specifically, we use a balanced class weighting approach to address the issue of class imbalance in the sampled data.

Then, based on the logistic regression prediction, we will determine whether the partial order relationship between the new sample point and the Pareto point changes (line 5).

TABLE I Configurable Parameters in the BOOM.

Module	Parameters	Candidates
FrontEnd	FetchWidth	4, 8
	FetchBufferEntry	8, 16, 24, 32
	RasEntry	16, 32, 36, 48
IDU	DecodeWidth	1, 2, 3, 4
	RobEntry	32, 64, 96, 128
	IntPhyRegister	52, 80, 100, 128
	FpPhyRegister	48, 64, 96, 128
ISU	Mem/Int/FpIssueWidth	1, 2, 3, 4
	Mem/Int/FpIssueEntry	8, 12, 16, 20, 24, 32, 40
LSU	LDQEntry	8, 16, 24
	STQEntry	8, 16, 24
ICache	ICacheWay	4, 8
	ICacheSet	32, 64
DCache	DCacheWay	4, 8
	DCacheSet	64, 128
	DCacheMSHR	2, 4, 8

If a change in the partial order relationship with a current Pareto point is detected (line 6), or if the new sample point is predicted to be a new Pareto point at high fidelity (line 11), we will proceed to the next fidelity level and obtain a more accurate evaluation report through the corresponding VLSI simulation process. Finally, through partial order prediction between the newly sampled point and the current Pareto points, we can effectively select the appropriate fidelity level and acquire multi-fidelity data.

IV. Experiment

A. Experimental Setup and Benchmarks

To evaluate the effectiveness of our proposed method, we conducted tests on two RISC-V CPU designs: BOOM with 20 design parameters and Rocket with 8 design parameters. Early architectural-level simulations are conducted using Gem5 [6], a cycle-accurate simulator, and McPAT [25], an integrated power and area modeling framework for architectures. Chipyard is used to generate RTL code from parameterized Chisel files. Subsequent PPA values for RTL and netlists are obtained from Synopsys VCS [26], Cadence Genus [27], and Cadence Joules [28] with the ASAP7 7nm PDK [29]. We use official RISC-V benckmarks [30] as workloads in the DSE. Multi-fidelity data encompasses simulation data generated at the architectural, RTL, and netlist levels. The multiple optimization objectives include power, area, and cycles. The cycles here are the average cycles of the benchmarks' inference performance on the CPU.

TABLE I presents the design space of the BOOM microarchitecture explored in our study. Hypervolume (HV) and average distance to reference set (ADRS) are used to compare the effectiveness of each multi-fidelity design space exploration method. Both metrics here are evaluated on the highest fidelity data learned by each method. To provide a comprehensive comparison of our method, we select several previous multi-fidelity algorithms and a classic single-fidelity algorithm:

- FPL'18 [12]: Method based on linear multi-fidelity and independent multi-objective models.

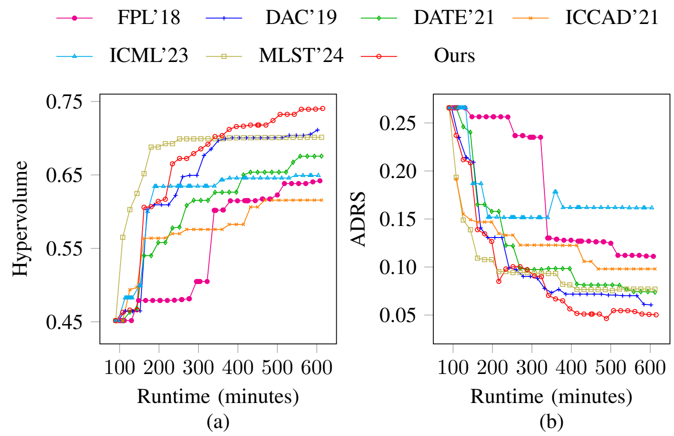

Fig. 4 Comparisons on the BOOM, (a) Hypervolume changes with EDA tool runtime. (b) ADRS changes with runtime.

- DAC'19 [13]: Method based on a nonlinear Gaussian model and a sequential fidelity strategy.
- DATE'21 [14]: Method with a correlated multiobjective non-linear optimization algorithm.
- ICCAD'21 [31]: Single-fidelity method with active learning-based initial dataset sampling and Gaussian process with deep kernel learning.
- ICML'23 [15]: Method utilizing one-step lookahead acquisition function based on the Knowledge Gradient.
- MLST'24 [16]: Method with trust-based multi-objective optimization.

Since verifying every sampled microarchitecture design online is time-consuming, we construct two offline datasets. Each design is simulated at the architecture, RTL, and netlist levels. The BOOM dataset contains 3153 parameter configurations, while the Rocket dataset contains 640 configurations. These two datasets consume 4,423 CPU hours to run the VLSI flow and obtain PPA values for each configuration at different fidelity levels. Our method and the baselines share the same randomly initialized parameter configurations. All experiments are conducted on a Linux server with Intel Xeon CPU (Platinum 8475B) and 512 GB RAM.

B. Comparisons against Previous Works

Considering the varying time costs of simulations at different fidelity levels, we initialize the models for all multi-fidelity methods using 30 architectural-level data, 15 RTL-level data, and 2 netlist-level data. Specifically, for the single-fidelity method [31], we initialize it with 6 high-fidelity data. Additionally, we replace the expected improvement acquisition function used in DAC'19 [13] with the EHVI acquisition function employed in this work to adapt it to our proposed multi-objective optimization problem. To compare the performance of different methods, we set the EDA tool runtime budget to 600 minutes.

TABLE II shows the comparison of our method with previous methods on two different RISC-V CPUs. The results show that our proposed multi-objective optimization method

TABLE II Comparisons against previous works.

Metric \ Method		FPL'18 [12]	DAC'19 [13]	DATE'21 [14]	ICCAD'21 [31]	ICML'23 [15]	MLST'24 [16]	**Ours**
BOOM	HV	0.6420 (100%)	0.7112 (↑10.8%)	0.6756 (↑ 5.2%)	0.6157 (↓4.1%)	0.6493 (↑ 1.1%)	0.7013 (↑ 9.2%)	**0.7405** (**↑15.3%**)
	ADRS	0.1111 (100%)	0.0606 (↓45.5%)	0.0738 (↓33.6%)	0.0980 (↓11.8%)	0.1616 (↑45.4%)	0.0770 (↓30.7%)	**0.0502** (**↓54.8%**)
Rocket	HV	0.6799 (100%)	0.7214 (↑ 6.1%)	0.7367 (↑ 8.4%)	0.6675 (↓ 1.8%)	0.6862 (↑ 0.9%)	0.7337 (↑ 7.9%)	**0.7513** (**↑10.5%**)
	ADRS	0.2185 (100%)	0.0963 (↓55.9%)	0.1408 (↓35.6%)	0.1745 (↓20.2%)	0.1011 (↓53.7%)	0.1547 (↓29.2%)	**0.0894** (**↓59.1%**)
Average	HV	0.6610 (100%)	0.7163 (↑ 8.4%)	0.7062 (↑ 6.8%)	0.6416 (↓ 2.9%)	0.6678 (↑ 1.0%)	0.7175 (↑ 8.5%)	**0.7459** (**↑12.9%**)
	ADRS	0.1648 (100%)	0.0785 (↓52.4%)	0.1073 (↓34.9%)	0.1363 (↓17.3%)	0.1313 (↓20.3%)	0.1159 (↓29.7%)	**0.0698** (**↓57.7%**)

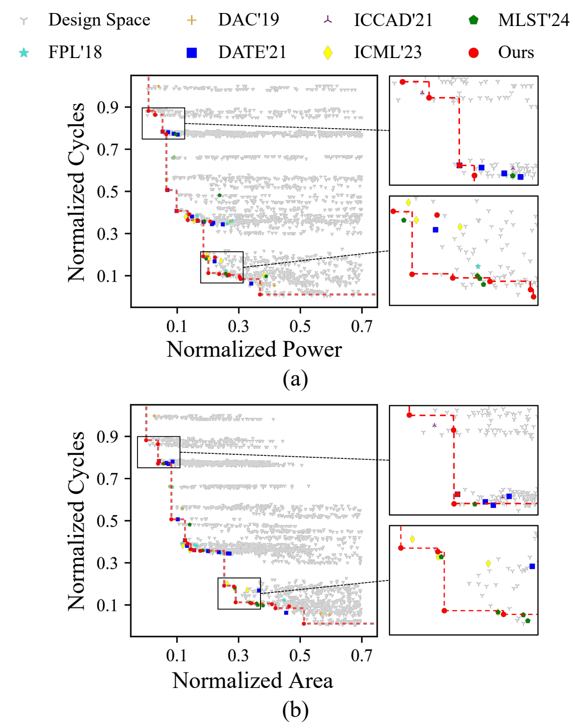

Fig. 5 Learned Pareto-optimal set of BOOM in cycles-power & cycles-area metric space. The red dashed line represents the Pareto frontier learned by our proposed method.

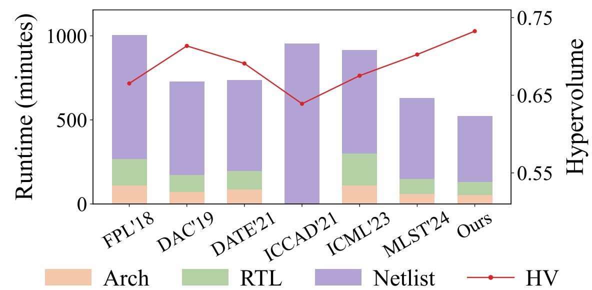

Fig. 6 Comparison of runtime distribution and hypervolume on the BOOM across different multi-fidelity optimization methods.

can better explore the search space under the same runtime allocation. Our method achieves the highest hypervolume and the lowest ADRS in the normalized space of the three metrics (cycles, power, and area). On average of hypervolume metrics, our method is 12.9% higher than FPL'18 [12], 4.5% higher than DAC'19 [13], 6.1% higher than DATE'21 [14], 10.0% higher than ICCAD'21 [31], 11.9% higher than ICML'23 [15], 4.4% higher than MLST'24 [16].

Fig. 4 shows the increase of hypervolume and the decrease of ADRS of each method on the BOOM dataset. The red curve rises faster than others in hypervolume. It falls faster than others in ADRS, which means that our method enables a more efficient and comprehensive exploration of the entire design space compared to other approaches. By leveraging low-fidelity data and partial order prediction, our method avoids spending excessive time on unexplored regions with suboptimal performance. Fig. 5 visually illustrates the distribution of the Pareto frontiers learned by various methods in the cycles-power and cycles-area spaces. It is evident that our method not only more effectively explores the Pareto frontier but also achieves a more comprehensive exploration of the entire design space.

Fig. 6 shows the distribution of simulation time across different fidelity levels consumed by our method and others during the exploration of the BOOM design space, along with the final hypervolume values achieved. As shown in the figure, compared to single-fidelity [31] and linear multi-fidelity methods [12], the non-linear multi-fidelity approach significantly enhances the efficiency of design space exploration. The figure also shows that on discrete fidelity data, our method, by constructing a corresponding GP model for each fidelity level, learns multi-fidelity data more effectively than using a single GP model to handle all fidelity levels [15], [16]. Guided by partial order prediction, our method significantly reduces the overall time cost of design space exploration.

V. Conclusion

In this paper, we have proposed an efficient multi-fidelity Bayesian optimization method for multi-objective optimization problems. We effectively fuse data from different fidelities by employing a non-linear multi-fidelity model to learn the complex correlation between low-fidelity and high-fidelity data. Under the inference of logistic regression function, we determine whether a more time-consuming but accurate evaluation is necessary by assessing whether the partial order relationship between two points changes across different fidelity levels. Experiments in exploring different RISC-V design spaces demonstrate the effectiveness and superior performance of the proposed framework.

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